UK Housing Market API 1.0.0 OAS 3.0

REST API for UK housing sales, affordability, and rental metrics based on ONS and UK House Price Index (HPI) data. Provides housing-sales data, buyer-dwelling breakdowns, affordability metrics, rental metrics, and an affordability index that compares your salary to regional rents and implied borrower income.

Servers

/ - Current host

Authorize

Lookups Reference data for filters and dropdowns

GET `/api/regions` List regions

Returns all regions (e.g. United Kingdom and any areas from rental data). Use for dropdowns and filtering other endpoints by region_id or region_name. No query parameters; returns the full list.

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	Success	*No links*

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
{
  "count": 0,
  "limit": 0,
  "offset": 0,
  "data": [
    {
      "id": 1,
      "name": "United Kingdom",
      "gss_code": "K02000001"
    }
  ]
}
```


500	Internal Server Error	*No links*

Media type

application/json

Example Value Schema

Code	Description	Links

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

POST `/api/regions` Create a region (admin)

Creates a new region. Requires API key (use Authorize button). Used by admins to add regions for rental or housing data.

Parameters Try it out

No parameters

Request body required — application/json

Example Value Schema

```
{
  "name": "London",
  "gss_code": "E12000007"
}
```

Responses

Code	Description	Links
201	Created. Returns the created region.	*No links*

Media type: application/json

Controls `Accept` header.

Example Value Schema

```
{
  "id": 1,
  "name": "United Kingdom",
  "gss_code": "K02000001"
}
```

400	Bad Request (e.g. name missing)	*No links*

Media type: application/json

Example Value Schema

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

401	Unauthorized (missing or invalid API key)	*No links*

Media type: application/json

Example Value Schema

```
{
  "error": "Bad Request",
```

Code	Description	Links

```
  "message": "salary query parameter is required"
}
```

409 Conflict (region with name or gss_code already exists) — *No links*

Media type: application/json

Example Value | Schema

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

500 Internal Server Error — *No links*

Media type: application/json

Example Value | Schema

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

PUT `/api/regions/{id}` Update a region (admin)

Updates an existing region by ID. Requires API key (use Authorize button). At least one of name or gss_code must be provided.

Parameters — Try it out

Name	Description
id * required `integer` *(path)*	Region ID to update. Path parameter; use the id from GET /api/regions. *Example* : 1 1

Request body required — application/json

Example Value | Schema

```
{
  "name": "string",
  "gss_code": "string"
}
```

Responses

Code	Description	Links
200	Success. Returns the updated region. Media type: application/json	*No links*

Code	Description	Links
	Controls `Accept` header.	

Example Value Schema

```
{
  "id": 1,
  "name": "United Kingdom",
  "gss_code": "K02000001"
}
```

400	Bad Request (invalid id or body)	*No links*

Media type

application/json

Example Value Schema

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

401	Unauthorized (missing or invalid API key)	*No links*

Media type

application/json

Example Value Schema

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

404	Not Found (region not found)	*No links*

Media type

application/json

Example Value Schema

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

409	Conflict (name or gss_code already in use)	*No links*

Media type

application/json

Example Value Schema

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

500	Internal Server Error	*No links*

Media type

application/json

Example Value Schema

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

DELETE `/api/regions/{id}` Delete a region (admin)

Deletes a region by ID. Requires API key (use Authorize button). Related data (housing sales, rental metrics, etc.) may be cascade-deleted.

Parameters Try it out

Name	Description
id * required integer *(path)*	Region ID to delete. Path parameter; use the id from GET /api/regions. *Example* : 2 2

Responses

Code	Description	Links
204	No Content. Region deleted.	*No links*

400	Bad Request (invalid id)
	Media type
	application/json
	Example Value Schema
	{
	"error": "Bad Request",
	"message": "salary query parameter is required"
	}
		No links

401	Unauthorized (missing or invalid API key)
	Media type
	application/json
	Example Value Schema
	{
	"error": "Bad Request",
	"message": "salary query parameter is required"
	}
		No links

404	Not Found (region not found)
	Media type
	application/json
	Example Value Schema
	{
	"error": "Bad Request",
	"message": "salary query parameter is required"
	}
		No links

500	Internal Server Error Media type application/json **Example Value** Schema { "error": "Bad Request",	*No links*

Code	Description	Links

```
  "message": "salary query parameter is required"
}
```

GET `/api/property-types` List property types

Returns property types (All property types, Detached, Semi-detached, Terraced, Flats and maisonettes). Use the returned id as property_type_id in GET /api/housing-sales.

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	Success. Data array contains PropertyType objects. Media type: application/json Controls `Accept` header. **Example Value** Schema	*No links*

```
{
  "count": 0,
  "data": [
    {
      "id": 0,
      "type_name": "string"
    }
  ]
}
```

500	Internal Server Error Media type: application/json **Example Value** Schema	*No links*

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

GET `/api/buyer-dwelling-categories` List buyer/dwelling categories

Returns buyer-dwelling categories (e.g. First time buyers, Former owner-occupiers, New dwellings, Other dwellings). Use the returned id as category_id in GET /api/housing-sales-by-buyer and GET /api/affordability.

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	Success. Data array contains BuyerDwellingCategory objects.	*No links*

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
{
  "count": 0,
  "data": [
    {
      "id": 0,
      "name": "string"
    }
  ]
}
```


500	Internal Server Error	*No links*

Media type

application/json

Example Value Schema


```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```


Housing & rental data Time-series and filtered datasets

GET `/api/housing-sales` Housing sales data (UKHPI)

Returns housing sales data from the UK House Price Index (UKHPI) with JOINs to regions and property types. Supports filtering by region, year, period range, property type, new/existing build, and price range. Paginated with limit/offset.

Parameters Try it out

Name	Description
region_id `integer` *(query)*	Filter by region ID. Get IDs from GET /api/regions. *Example* : 1 1
region_name `string` *(query)*	Filter by region name (case-insensitive). Use instead of region_id if preferred. *Example* : United Kingdom United Kingdom
year `integer` *(query)*	Filter by calendar year. Valid range 1900–2100. Sets period_from/period_to to that year. *Example* : 2025 2025
period_from	Start of period (YYYY-MM-DD). Use with period_to for a date range.

Name	Description
string($date) (query)	*Example* : 2025-01-01 2025-01-01
period_to string($date) (query)	End of period (YYYY-MM-DD). *Example* : 2025-12-31 2025-12-31
property_type_id integer (query)	Filter by property type ID. Get IDs from GET /api/property-types. *Example* : 1 1
is_new_build boolean (query)	Filter by new build (true) or existing (false). *Example* : false false
min_price number (query)	Minimum average price (£). Must be non-negative; must be ≤ max_price if both set. *Example* : 100000 100000
max_price number (query)	Maximum average price (£). Must be non-negative. *Example* : 500000 500000
limit integer (query)	Max number of rows to return. Default 100, max 500. *Default value* : 100 *Example* : 100 100
offset integer (query)	Number of rows to skip (pagination). Non-negative integer. *Default value* : 0 *Example* : 0 0

Responses

Code	Description	Links
200	Success. Data contains housing sales rows with region_name, property_type_name, period, avg_price, etc. Media type application/json Controls `Accept` header.	*No links*

Code	Description	Links

Example Value Schema

```
{
  "count": 0,
  "limit": 0,
  "offset": 0,
  "data": [
    "string"
  ]
}
```

400 Bad Request (invalid parameters) *No links*

Media type

application/json

Example Value Schema

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

500 Internal Server Error *No links*

Media type

application/json

Example Value Schema

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

GET `/api/housing-sales-by-buyer` Housing sales by buyer and dwelling type

Returns average house prices by buyer type (e.g. First time buyers, Former owner-occupiers) and dwelling type (New/Other/All), with JOINs to regions and buyer-dwelling categories. Data sourced from ONS Table 28. Filter by region, period, category, and price range.

Parameters Try it out

Name	Description
region_id *integer* *(query)*	Filter by region ID. Get IDs from GET /api/regions. *Example* : 1 1
region_name *string* *(query)*	Filter by region name (case-insensitive). *Example* : United Kingdom United Kingdom
year *integer* *(query)*	Filter by year (1900–2100). *Example* : 2025 2025

Name	Description
period_from string($date) (query)	Start date YYYY-MM-DD. *Example* : 2025-01-01 2025-01-01
period_to string($date) (query)	End date YYYY-MM-DD. *Example* : 2025-12-31 2025-12-31
category_id integer (query)	Filter by buyer-dwelling category ID. Get IDs from GET /api/buyer-dwelling-categories. *Example* : 1 1
min_price number (query)	Minimum average price (£). Non-negative. *Example* : 100000 100000
max_price number (query)	Maximum average price (£). Non-negative; must be ≥ min_price if both set. *Example* : 400000 400000
limit integer (query)	Max rows to return. Default 100, max 500. *Default value* : 100 *Example* : 100 100
offset integer (query)	Pagination offset. Non-negative. *Default value* : 0 *Example* : 0 0

Responses

Code	Description	Links
200	Success. Data contains rows with region_name, category_name, period, avg_price. Media type application/json Controls `Accept` header. **Example Value** Schema	*No links*

```
{
  "count": 0,
```

Code	Description	Links

```
  "limit": 0,
  "offset": 0,
  "data": [
    "string"
  ]
}
```

400 Bad Request — *No links*

Media type: application/json

Example Value Schema

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

500 Internal Server Error — *No links*

Media type: application/json

Example Value Schema

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

GET /api/rental-metrics Rental metrics

Returns rental price metrics by region and period: all-property rent, annual change %, and breakdowns (one bed, four+ bed, detached, terraced, flat/maisonette). Filter by region, period, and rent range.

Parameters — Try it out

Name	Description
region_id integer *(query)*	Filter by region ID. *Example* : 1 1
region_name string *(query)*	Filter by region name. *Example* : United Kingdom United Kingdom
year integer *(query)*	Filter by year (1900–2100). *Example* : 2025 2025
period_from string($date) *(query)*	Start date YYYY-MM-DD. *Example* : 2025-01-01 2025-01-01

Name	Description
period_to string($date) (query)	End date YYYY-MM-DD. *Example* : 2025-12-31 2025-12-31
min_rent number (query)	Minimum rental price (£/month). Non-negative; must be ≤ max_rent if both set. *Example* : 500 500
max_rent number (query)	Maximum rental price (£/month). Non-negative. *Example* : 2000 2000
limit integer (query)	Max rows. Default 100, max 500. *Default value* : 100 *Example* : 100 100
offset integer (query)	Pagination offset. *Default value* : 0 *Example* : 0 0

Responses

Code	Description	Links
200	Success. Data contains rental metrics (rental_price_all, annual_change_pct, etc.). Media type application/json Controls `Accept` header. **Example Value** Schema	*No links*

```
{
  "count": 0,
  "limit": 0,
  "offset": 0,
  "data": [
    "string"
  ]
}
```

400	Bad Request Media type application/json **Example Value** Schema	*No links*

Code	Description	Links

	{
	"error": "Bad Request",
	"message": "salary query parameter is required"
	}	

500	Internal Server Error
	Media type
	application/json
	Example Value Schema
	{
	"error": "Bad Request",
	"message": "salary query parameter is required"
	}
		No links

Affordability Affordability metrics and index

GET `/api/affordability` Affordability metrics

Returns ONS affordability metrics (advance price %, price/income ratio, advance/income ratio) by region and buyer-dwelling category. Data from ONS Table 30. Filter by region, period, and category.

Parameters Try it out

Name	Description
region_id integer *(query)*	Filter by region ID. *Example* : 1 1
region_name string *(query)*	Filter by region name. *Example* : United Kingdom United Kingdom
year integer *(query)*	Filter by year (1900–2100). *Example* : 2025 2025
period_from string($date) *(query)*	Start date YYYY-MM-DD. *Example* : 2025-01-01 2025-01-01
period_to string($date) *(query)*	End date YYYY-MM-DD. *Example* : 2025-12-31 2025-12-31
category_id integer	Filter by buyer-dwelling category ID. *Example* : 1

Name	Description
(query)	1
limit integer (query)	Max rows. Default 100, max 500. *Default value* : 100 *Example* : 100 100
offset integer (query)	Pagination offset. *Default value* : 0 *Example* : 0 0

Responses

Code	Description	Links
200	Success. Data contains affordability metrics (advance_price_pct, price_income_ratio, etc.). Media type application/json Controls `Accept` header. **Example Value** Schema	*No links*

```
{
  "count": 0,
  "limit": 0,
  "offset": 0,
  "data": [
    "string"
  ]
}
```

400	Bad Request Media type application/json **Example Value** Schema	*No links*

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

500	Internal Server Error Media type application/json **Example Value** Schema	*No links*

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

GET `/api/affordability-index` Affordability index (salary vs rent)

Compares your annual salary to the latest regional average rent and implied borrower income. Returns rent-to-salary ratios and a health score (Affordable ≤30%, Stretched ≤40%, Unaffordable >40%). Uses latest rental_metrics and regions for the chosen region.

Parameters

Try it out

Name	Description
salary * required number *(query)*	Your annual salary (required). Must be a positive number. 35000
region_id integer *(query)*	Region ID (optional). If omitted, region_name or default United Kingdom is used. region_id
region_name string *(query)*	Region name (optional string). Used if region_id is not provided. Defaults to United Kingdom if neither is set. United Kingdom

Responses

Code	Description	Links
200	Success. Returns region, inputs, rental snapshot, ratios, health score, and explanation.	*No links*

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
{
  "region": {
    "id": 0,
    "name": "string",
    "gss_code": "string",
    "implied_avg_borrower_income": 0,
    "implied_income_year": "string"
  },
  "inputs": {
    "annual_salary": 0
  },
  "rental_snapshot": {
    "period": "string",
    "monthly_rent_all_property_types": 0,
    "annual_rent_all_property_types": 0
  },
  "ratios": {
    "rent_to_salary_ratio": 0,
    "rent_to_salary_percent": 0,
    "rent_to_region_income_percent": 0
  },
  "health_score": "Affordable",
  "health_explanation": "string"
}
```


400	Bad Request (e.g. missing or invalid salary)	*No links*

Media type

application/json

Example Value Schema

Code	Description	Links

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

401 Unauthorized (invalid or missing API key for write operations; GET is public) — *No links*

Media type: application/json

Example Value Schema

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

404 Not Found (region or rental metrics not found) — *No links*

Media type: application/json

Example Value Schema

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

500 Internal Server Error — *No links*

Media type: application/json

Example Value Schema

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

AI insights

LLM-generated market analysis (deployer-configured Gemini; no consumer API key)

GET `/api/analysis/market-summary` Market summary by query (LLM)

Expert market analysis: retrieves latest rental, last 6 months HPI, and price-to-income ratio for the region, feeds them to Gemini, and returns a 150-word executive summary. Optional tailoring: salary (personalise rent affordability), focus (first_time_buyer / investor / rent_vs_buy), property_type_id (HPI for that property type). Region: region_id or region_name (default United Kingdom). Deployer sets GEMINI_API_KEY.

Parameters Try it out

Name	Description
region_id integer *(query)*	Region ID (optional). region_id
region_name string *(query)*	Region name (optional). Used if region_id not set; defaults to United Kingdom. United Kingdom
salary number *(query)*	User annual salary in GBP. Tailors summary with rent-as-%-of-salary and affordability angle. 35000

Name	Description
focus string *(query)*	Scenario focus: first-time buyer affordability, investor outlook, or rent vs buy. *Available values* : first_time_buyer, investor, rent_vs_buy --
property_type_id integer *(query)*	Property type ID for HPI data (e.g. Detached, All property types). Use GET /api/property-types for IDs. property_type_id

Responses

Code	Description	Links
200	Success. Returns region, summary, data_snapshot (includes user inputs when provided), generated_at. Media type application/json Controls Accept header. **Example Value** Schema	*No links*


```
{
  "region": {
    "id": 0,
    "name": "string",
    "gss_code": "string"
  },
  "summary": "string",
  "data_snapshot": {
    "region_name": "string",
    "latest_rent_period": "string",
    "monthly_rent_gbp": 0,
    "annual_rent_gbp": 0,
    "rent_annual_change_pct": 0,
    "implied_avg_borrower_income_gbp": 0,
    "implied_income_year": "string",
    "avg_house_price_gbp": 0,
    "house_price_period": "string",
    "house_price_annual_change_pct": 0,
    "hpi_months_included": 0,
    "price_to_income_ratio": 0,
    "price_to_income_period": "string",
    "user_salary_gbp": 0,
    "rent_to_salary_pct": 0,
    "focus": "first_time_buyer",
    "property_type_id": 0,
    "property_type_name": "string"
```


400	Bad Request (e.g. invalid property_type_id) Media type application/json **Example Value** Schema	*No links*


```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```


404	Region or rental metrics not found Media type application/json **Example Value** Schema	*No links*


```
{
  "error": "Bad Request",
```

Code	Description	Links

```
  "message": "salary query parameter is required"
}
```

500 Internal Server Error — *No links*

Media type: application/json

Example Value | Schema

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

502 AI service error or empty response — *No links*

Media type: application/json

Example Value | Schema

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

503 AI insights not configured (GEMINI_API_KEY not set) — *No links*

Media type: application/json

Example Value | Schema

```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

GET `/api/analysis/{region_id}` Market summary by region ID (LLM)

Same as GET /api/analysis/market-summary but with region_id in the path. Optional query params: salary, focus, property_type_id for user-tailored analysis.

Parameters — Try it out

Name	Description
region_id * required integer *(path)*	Region ID. region_id
salary number *(query)*	User annual salary in GBP. Tailors summary with rent-as-%-of-salary. 35000
focus string *(query)*	Scenario: first_time_buyer, investor, or rent_vs_buy. *Available values* : first_time_buyer, investor, rent_vs_buy --
property_type_id integer *(query)*	Property type ID for HPI data. Use GET /api/property-types. property_type_id

Responses

Code	Description	Links
200	Success. Returns region, summary, data_snapshot, generated_at.	*No links*

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
{
  "region": {
    "id": 0,
    "name": "string",
    "gss_code": "string"
  },
  "summary": "string",
  "data_snapshot": {
    "region_name": "string",
    "latest_rent_period": "string",
    "monthly_rent_gbp": 0,
    "annual_rent_gbp": 0,
    "rent_annual_change_pct": 0,
    "implied_avg_borrower_income_gbp": 0,
    "implied_income_year": "string",
    "avg_house_price_gbp": 0,
    "house_price_period": "string",
    "house_price_annual_change_pct": 0,
    "hpi_months_included": 0,
    "price_to_income_ratio": 0,
    "price_to_income_period": "string",
    "user_salary_gbp": 0,
    "rent_to_salary_pct": 0,
    "focus": "first_time_buyer",
    "property_type_id": 0,
    "property_type_name": "string"
```


400	Bad Request (e.g. invalid property_type_id)	*No links*

Media type

application/json

Example Value Schema


```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```


404	Region or rental metrics not found	*No links*

Media type

application/json

Example Value Schema


```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```


500	Internal Server Error	*No links*

Media type

application/json

Example Value Schema


```
{
  "error": "Bad Request",
  "message": "salary query parameter is required"
}
```

Code	Description	Links
502	AI service error or empty response Media type application/json **Example Value** Schema `{ "error": "Bad Request", "message": "salary query parameter is required" }`	*No links*
503	AI insights not configured (GEMINI_API_KEY not set) Media type application/json **Example Value** Schema `{ "error": "Bad Request", "message": "salary query parameter is required" }`	*No links*

Health API health check

GET `/api/names` API health check

Simple health check that verifies the API and database connection. Returns a success message and server timestamp. No parameters required.

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	Success. API and database are reachable. Media type application/json Controls `Accept` header. **Example Value** Schema `{ "message": "string", "timestamp": "string" }`	*No links*
500	Internal Server Error (e.g. database unreachable) Media type application/json **Example Value** Schema `{ "error": "Bad Request",`	*No links*

Code	Description	Links


```
  "message": "salary query parameter is required"
}
```


Schemas

Error

SuccessList

Region

RegionCreate

RegionUpdate

PropertyType

BuyerDwellingCategory

AffordabilityIndexSuccess

HealthCheckResponse

MarketSummarySuccess